MACHINE GUARDIAN · MAC-ATTACK

Machine Guardian — Training Run 3 and Expanded Dataset Evaluation

Seven Paderborn development bearings (four new), nested leave-one-bearing-out validation, five training configurations, INT8 and RTL verification of the trained 8 → 32 → 16 → 2 multi-task MLP

7
development bearings, 4 new (B08, B09, B11, B12)

6 of 7
bearings alarmed before the stop (B12 missed)

1.78 µs
per inference, trained network on the RTL

MAC-Attack senior capstone · Machine Guardian
Prepared with Claude Code · 3 November 2026
Source: model/results/supervised/run3/ (prep, train, hardware, supplement, analysis)
The output is a deterioration score, not a validated probability of physical failure.

Contents

01 Executive Summary

02 Project Objectives and Current Architecture

03 Summary of Training Runs 1 and 2

04 New Dataset Integration (B08, B09, B11, B12)

05 Dataset Statistics and Label Distribution

06 Feature Extraction and Preprocessing

07 Training Run 3 Configuration

08 Improvements and Changes from Run 2

09 Training Results and Learning Curves

10 Per-Bearing Validation Results

11 Accuracy, Precision, Recall, F1 and Confusion Matrices

12 Early Warning and Alarm-Timing Comparison

13 Run 1 vs. Run 2 vs. Run 3 Comparison

14 Comparison Against the Original Anomaly Detector

15 Prediction Confidence, Calibration and Input-Clipping Analysis

16 INT8 Quantization and RTL Verification

17 Current Limitations and Uncertainties

18 Recommended Improvements for Training Run 4

19 FPGA Deployment Readiness

20 Final Conclusion

MAC-Attack senior capstone. Deterioration head of the 8 → 32 → 16 → 2 multi-task MLP, seven Paderborn development bearings, nested leave-one-bearing-out validation, INT8 and RTL verification. Prepared with Claude Code, 3 November 2026. The tables are generated by `model/report/run3_tables.py` from the saved results in `model/results/supervised/run3/`; the numbers in the text were taken by hand from those results and from `run1/`, `run2/` and `results/paderborn/`, and every one was checked by an independent fact-check against the files (section 20 lists them).

1. Executive Summary

What Run 3 set out to do

Runs 1 and 2 trained the network on only three bearings (B02–B04). Run 3 adds four more development bearings from the Paderborn run-to-failure dataset (B08, B09, B11, B12) and retrains the same 850-parameter network with five candidate training configurations. The candidates were compared with a nested, bearing-separated protocol, so that the selection can be judged on bearings it never saw.

Main results

Question	Answer
New datasets processed	**B08, B09, B11, B12**: 6,360 vibration files (14.9 GB, 13.9 GiB), every file CRC-checked, every hard data check passed, all four logs transcribed
Training data	7 bearings, **9,003 operating measurements**: 700 healthy-reference labels, 35 provisional end-stage labels, 8,268 unlabelled
Experiments	5 candidates × 7 outer folds × 5 seeds, plus 6 inner folds each: **1,225 trainings**, plus 10 final models and 3 reproduction checks
Selected configuration	**E**: weight decay 0.01, log-compressed inputs instead of the ±16 clip, temperature input fed as 0. Chosen by a rule declared before training, applied to bearing-separated results. The choice was decided by **2 alarm transitions in 35 runs**: E and **C** (the same with temperature) are statistically indistinguishable
Accuracy / precision / recall / F1 (latched alarm, labelled held-out rows)	0.9932 / 1.0000 / 0.8571 / 0.9231, the same for every candidate and for the nested selection. These row-level numbers are weak evidence: a rule that never alarms scores accuracy 0.952, and a rule that simply latches an alarm right after commissioning scores 1.000 (section 11)
Bearing-level outcome (the independent unit)	end stage latched before the stop on **6 of 7 bearings** (95 % interval 0.42–1.00). On the four unbiased new bearings: **3 of 4** (detector 4 of 4)
Healthy-reference false alarms	**0** of 3,500 healthy-row checks (7 bearings × 100 rows × 5 seeds)
End-stage recall	**6 of 7 bearings (85.7 %)**. Every MLP candidate misses **B12** entirely
Alarm stability	the unlatched (3-in-a-row) alarm's state changes fall from 30.3 (Run 1–2 settings) to **2.8** per run on average; the latched alarm changes at most once (once on 6 of 7 bearings, never on B12)
Earlier than the anomaly detector?	**No.** On every bearing the anomaly detector alarms at the same time as the selected network or earlier: by 0.7–1.0 min on B02, 3.9 min on B03, 1.0 min on B04, 210 min on B08, 6–8 min on B09 and 1.6–1.8 min on B11. It also catches B12 (40.8 min before the end), which the MLP never does
INT8 and RTL	**pass** after one export fix. INT8 changes 13 of 45,015 decisions for the 35 held-out E models (0.03 %, 3 on labelled rows) and 39 of 45,015 for the 5 final E models (0.09 %, 4 on labelled rows). 16 RTL simulations of trained networks are bit-exact: 382 cycles = 1.78 µs of accelerator compute per measurement. **Simulation only, not on the ZedBoard**
Outputs	deterioration score: **trained**. Motor Health Index: **untrained and frozen**

> **Key finding:** Run 3 makes the network much **steadier**, through log-compressed inputs and weight decay, and brings its first alarm close to the anomaly detector's on five of seven bearings (it comes about 3.5 h later on B08 and never on B12). Compared with Runs 1–2 on the same three bearings, the first alarm moves from 7.4 to 16.5 minutes before the end on B03, and the unlatched alarm's flicker on B04 drops from 43–47 changes to 1.

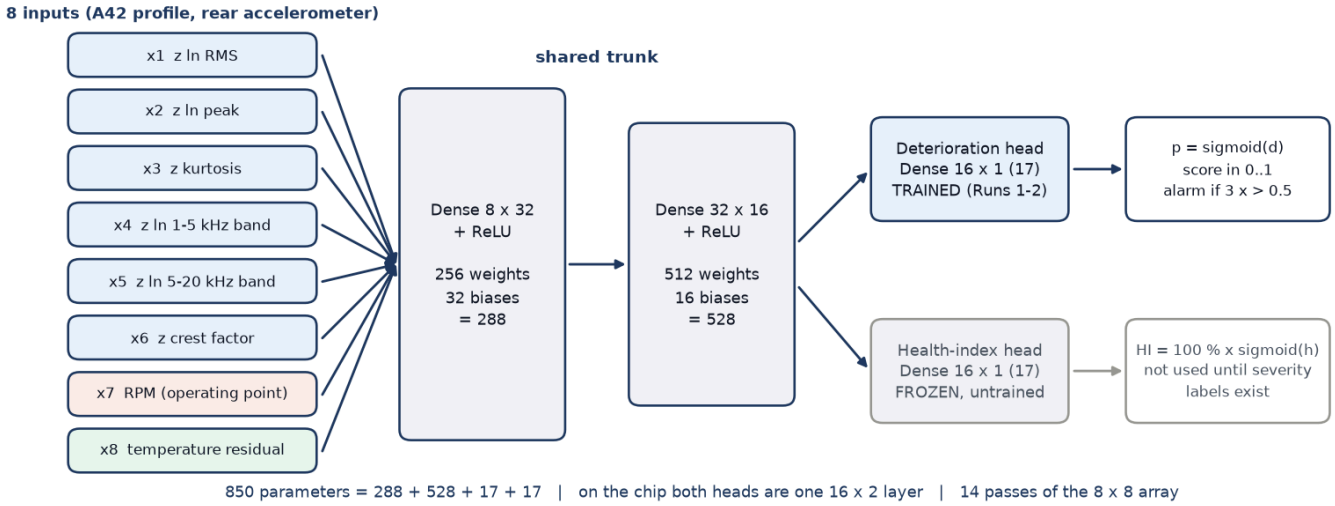
Key finding: the network **still does not outperform** the label-free anomaly detector. The likely reason, not yet tested by a dedicated experiment, lies in the labels and in B12's data. Every end-stage label on the other six bearings is an extreme, worn-out measurement (the largest vibration z-input is 18–51 standard deviations), so the network probably learns that only extreme deviation means damage. B12's end stage is only 2.5–3.2 standard deviations from its own commissioning on the rear sensor: the spread is inflated by a vibration step at M0061, and its stop was probably triggered by the front sensor. The network never alarms on B12.

Caution: with 7 bearings, each with 100 healthy rows from its own commissioning and 5 end-stage rows, the labelled metrics are largely met by construction. The healthy rows are the rows the inputs are normalised on, and the end-stage rows are extreme. The evidence that matters is per bearing: whether and when the alarm comes, compared with the anomaly detector, on bearings that played no part in the design (B08, B09, B11, B12).

Caution: "minutes before the last record" is the time before the experiment was stopped. It is **not** a verified time before physical failure or before deterioration began; neither is known. The score is a detector output, not a calibrated probability of failure.

2. Project Objectives and Current Architecture

Machine Guardian monitors a rotating machine's bearing and raises a warning when it starts to deteriorate. A small neural network is designed to run in INT8 on our own 8 × 8 systolic-array accelerator on the ZedBoard (Zynq XC7Z020); so far the trained networks have run only in RTL simulation, not on the board. The ARM processor extracts the features and runs the alarm logic.



Part	Run 3
Inputs	8 per 1.6-s measurement: 6 rear-accelerometer features (ln RMS, ln peak, kurtosis, ln 1–5 kHz and ln 5–20 kHz band energy, crest factor), shaft speed, temperature residual. The six vibration features (speed-corrected) and the speed are z-scores against the bearing's own first 100 operating measurements (its commissioning); the temperature residual comes from a per-bearing thermal model
Network	**unchanged**: 8 → 32 → 16 → 2, ReLU, 850 parameters (288 + 528 + 17 + 17)
Output 1	deterioration score p = sigmoid(logit), alarm after 3 consecutive p > 0.5, **latched** (Run 2). **Trained**
Output 2	Motor Health Index, HI = 100 % × sigmoid(h). **Untrained**, frozen at its initial weights and checked bit-identical after every training, because no measured severity labels exist
Hardware	INT8 weights, INT32 biases, 22-bit accumulators, 14 passes of the 8 × 8 array, both heads in one 16 × 2 last layer

3. Summary of Training Runs 1 and 2

	Run 1	Run 2
Bearings	B02, B03, B04 (3 folds, train on 2)	the same
Training	Adam, lr 0.001, 2,000 steps, batch 64 (half per class), temperature dropout 50 %, no weight decay, seeds 0–2	identical networks (byte-identical weights)
Alarm	on while the last 3 scores are above 0.5	**latched** after 3 above 0.5
Result	0 % healthy rows flagged and 100 % end-stage rows alarmed on every bearing, but the alarm flickered (up to 47 state changes) and came later than the anomaly detector	flicker removed (1 change), first alarm unchanged; scores shown to be over-confident (87 % below 0.001 or above 0.999); INT8 and RTL of a trained network passed
Open questions	—	does weight decay 0.01 generalise (it was found on the same 3 bearings)? does the ±16 clip hide information? does the temperature input's "not yet calibrated = 0" shortcut matter?

4. New Dataset Integration (B08, B09, B11, B12)

4.1 What was received and how it was integrated

Step	What was done	Result
Locate	five archives in `~/Downloads`: `B08.zip`, `B09_part1.zip`, `B09_part2.zip`, `B11.zip`, `B12.zip`	all found; nothing downloaded again
Read	the macOS `unzip` and `bsdtar` reported `B08.zip` as damaged ("start of central directory not found")	it is a ZIP64 archive over 4 GB, which those tools cannot read; Python's `zipfile` reads it (1,831 entries)
Extract	`extract_paderborn_zips.py` (new): checks every member path, refuses to overwrite, verifies each file's CRC-32 while streaming, records a SHA-256 manifest, merges B09's two parts	`model/newdata/<EXP>/` with `<EXP>_log.pdf`, two CSVs and `vibrationData/`
Count	files against the dataset description	B08 1,827, B09 1,813 (900 + 913), B11 1,953, B12 767: all complete
Logs	the four `<EXP>_log.pdf` transcribed into `paderborn_experiments.json` (previous version backed up); every other entry checked unchanged	start, warm-up, stop and stop reason per bearing (table 4.2)
Code fix	the new archives name files `data_B09_M00001.mat` (5 digits); `paderborn.py` accepted only 4	one-line change to accept 4 or 5 digits; B01–B04 results unchanged
Process	`paderborn.py B12 B11 B09 B08` (unchanged pipeline, 28 s)	`results/paderborn/<EXP>/` with processed CSV, `checks.json`, overview figure
Verify	an independent review by separate agents of every log transcription (against the PDF page images) and of the data, with each finding adversarially re-checked against the raw files	transcriptions correct; findings in table 4.4

4.2 The experiment logs

Bearing	Logged start	Warm-up	Logged stop	Stop reason (log)	Defect at dismantling	Regime
B08	04-02-2022 14:43	10 min (750 rpm, 2,500 N)	05-02-2022 01:20	temperature limit exceeded	inner race, ball	Gaussian load, 128 kHz
B09	18-02-2022 11:37	10 min	19-02-2022 03:59	accelerometer limit exceeded	outer race, ball	Gaussian load, 128 kHz
B11	17-08-2022 09:29	**none**	17-08-2022 18:17	accelerometer limit exceeded	inner race, outer race, ball	Gaussian load, 64 kHz, variable speed
B12	24-08-2022 08:51	**none**	24-08-2022 11:22	accelerometer limit exceeded	inner race, ball	Gaussian load, 64 kHz, variable speed

No log states the limit's value or which accelerometer triggered the stop.

4.3 Sampling frequencies and operating regimes

- **Sampling rates.** B08 and B09 are sampled at 128 kHz, B11 and B12 at 64 kHz, matching the description. `paderborn.py` detects the rate per file from its time vector. At 64 kHz the 20–64 kHz band cannot be computed in full (its upper part lies above the 32 kHz Nyquist frequency) and is left empty, and the 42.67-kHz signal is resampled by 2/3, so the A42 profile the model uses is identical in definition for both rates.
- **Operating regimes.** All four new bearings run under Gaussian random dynamic load, where Runs 1–2 had only sinusoidal load. B11 and B12 run at constant static (3,000 N) and dynamic (1,000 N) load with variable speed. Every speed set point lies inside the logged range; B09's recorded dynamic load does not (table 4.4).

- **The prototype anomaly detector cannot run on B11 and B12.** It uses the 20–64 kHz band, so every score is NaN at 64 kHz. The same detector on the 42.67-kHz features (A, F42, speed) runs on all seven bearings and is the common reference in this report.

4.4 Data-quality findings (documented, not repaired)

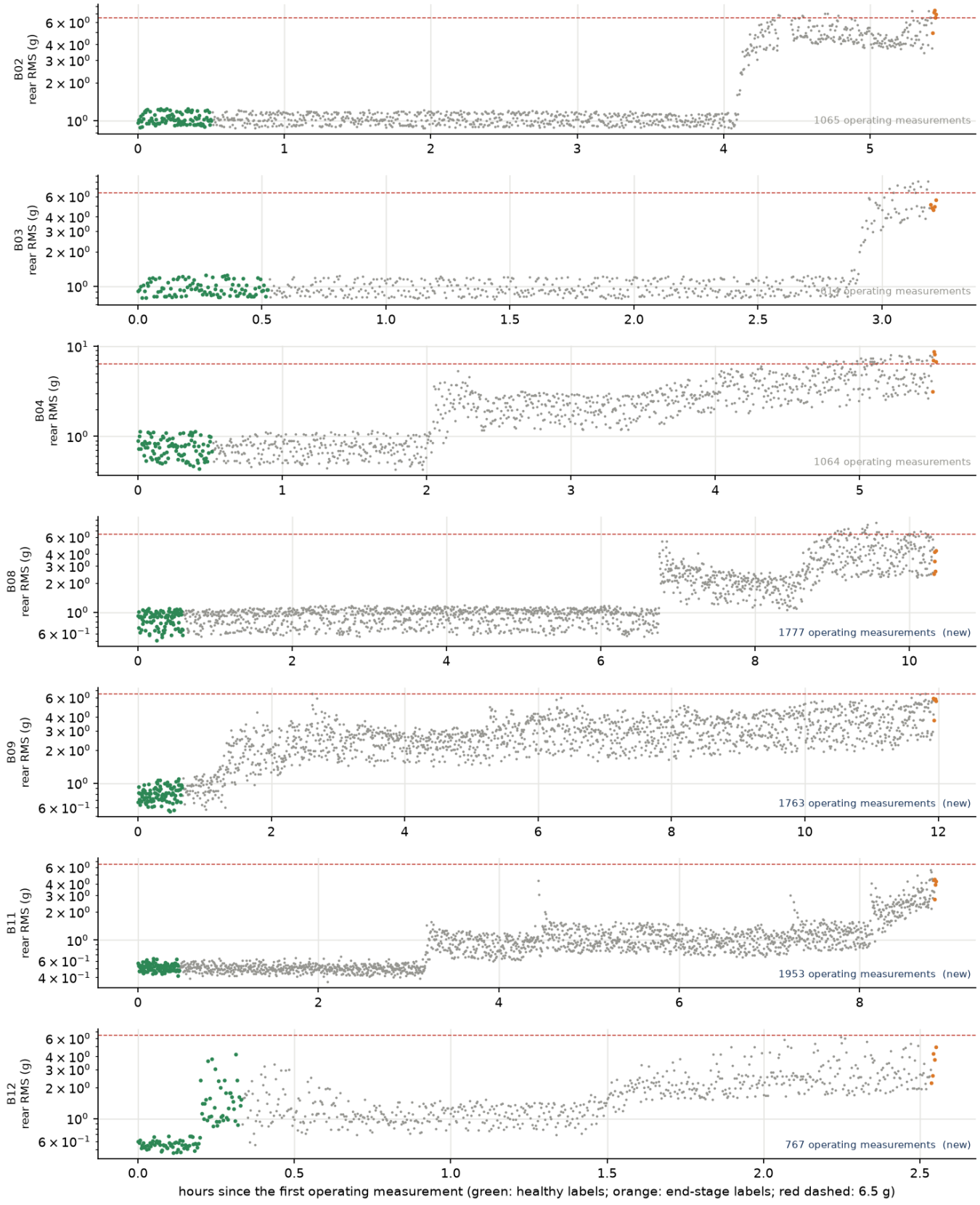
Nothing was discarded or silently corrected. Each finding was reproduced from the raw files by an independent check.

Bearing	Finding	Severity	Effect and decision
B12	**Undocumented, lasting vibration step at M0061** (12 min after the start): rear RMS rises from about 0.56 g to 1.37 g and stays there. It lies inside the healthy window and commissioning reference M0001–M0100	major	The healthy labels follow the rule and were kept. The step inflates B12's reference spread and shrinks its z-inputs (section 15.4). Reported as one of two reasons B12 is missed, with the weak rear-sensor signature
B12	The rear accelerometer never exceeds 6.2 g; the front reaches 8.5 g at the end. The stop was probably triggered by the front sensor, which the model does not use	major	End-stage labels kept (documented stop plus damage found at dismantling). Their rear-sensor signature is weak: the largest vibration z on the end-stage rows is 2.5–3.2
B11	Rear RMS peaks at 5.60 g; the front exceeds 6.5 g on 14 rows from M1875. The same front-sensor pattern	minor	as for B12; B02's 6.5 g limit, applied to the rear sensor, never fires on B11
B11	**The T1 temperature sensor loses contact twice** (from M0979) and then reads a flat ~43 °C while T2 keeps tracking the bearing	major	Corrupts the temperature input x8 on B11's later rows, including the end stage. A reason to prefer candidates without x8 (D, E)
B11, B12	No warm-up phase (log); the healthy window is the thermal run-in from a cold start	info	Temperature residual grows large after calibration (raw x8 up to +25 σ on B11 before its T1 fault, about +31 σ on B12 to the end); B11's −60 σ at the end comes from the T1 fault
B09	**The recording starts at 15:51:44, 4 h 14 min after the logged start (11:37)**. The documented warm-up is at the start of the recording and the end matches the log	minor	Nothing depends on the logged start; documented
B09	The recorded set dynamic load reaches 1,500 N (78.6 % of operating rows above 1,000 N), against the log's range of 750–1,000 N	major (documentation)	Load is not a model input; log and data disagree; documented
B09	The rear sensor exceeds 6.5 g on one row only (6.55 g at M1790; the end-stage rows reach at most 5.92 g), so B02's 6.5 g limit (3 rows in a row) never fires	minor	Limit value not documented
B09	The T2 sensor is implausible from 17:35 (spikes to 168 °C)	minor	T2 is not used (x8 uses T1)
B09	Rear-sensor kurtosis in the healthy window is high (median 4.67 against about 3.1 on the other bearings), and vibration leaves the commissioning level within the first hour	minor	The anomaly detector alarms at M0207–M0211, very early in a 12-h test; whether this is real damage cannot be verified
B08	An undocumented 430-s stop between the warm-up and operation (14:53:24–15:00:34)	minor	The healthy window lies after it. With the stop, the thermal calibration fails its validity rule (V1, and V3 at 1.31 °C), so **x8 = 0 for B08**
B08, B09, B11	The measured static load is systematically below the set value (B08 by 96 N on average; 729, 962 and 162 rows more than 150 N off)	minor	A property of the rig; load is not an input
B02–B04, B08, B09, B11	The measurement interval during operation is 16–23 s (median), longer than the ~12 s pause stated on page 1 of the logs; B12 (12 s) and the warm-up rows match it	info	No effect
all	`checks.json` key `combined_set_at_4300` counts rows at 4,000 N for these bearings	cosmetic	Key name only

5. Dataset Statistics and Label Distribution

Bearing	Sampling	Files	Warm-up rows	Operating rows	Segments	Recorded span	Healthy (0)	End-stage (1)	Unlabelled	Temperature input	Rows with an input clipped at ±16
B02	128 kHz	1116	51	1065	2	5.6 h	M0052–M0151	M1112–M1116	960	valid	252
B03	128 kHz	614	0	614	1	3.2 h	M0001–M0100	M0610–M0614	509	valid	62
B04	128 kHz	1114	50	1064	1	5.7 h	M0051–M0150	M1110–M1114	959	valid	672
B08 (new)	128 kHz	1827	50	1777	2	10.6 h	M0051–M0150	M1823–M1827	1672	masked (x8 = 0)	607
B09 (new)	128 kHz	1813	50	1763	1	12.1 h	M0051–M0150	M1809–M1813	1658	valid	517
B11 (new)	64 kHz	1953	0	1953	1	8.8 h	M0001–M0100	M1949–M1953	1848	valid (V1–V3), but T1 sensor fault from M0979	1247
B12 (new)	64 kHz	767	0	767	1	2.5 h	M0001–M0100	M0763–M0767	662	valid	438

Totals for the seven training bearings: 9,003 operating measurements, 700 healthy-reference labels, 35 provisional end-stage labels and 8,268 unlabelled. Counting B01 (processed, no labels), 9,581 files have now been processed.



5.1 How the labels were applied, and checked

The labelling procedure is unchanged (`labels.py`, rule fixed on 2 Nov 2026):

- **Healthy (0):** the first 100 operating measurements of a new bearing.
- **Provisional end-stage (1):** the last 5 operating measurements, and only if the recording ends within 120 s of the logged stop and those rows contain no stop.

- **Everything in between:** unlabelled.

Before the labels were applied, each new bearing was checked:

Check	B08	B09	B11	B12
Stop documented (log + description)	temperature	vibration	vibration	vibration
Last record vs logged stop (the log gives minutes only)	+58 s	+44 s	+1 s	+23 s
Records after the logged minute (stop + 59 s)	0	0	0	0
Last 5 rows in one segment	yes	yes	yes	yes
Damage documented after dismantling	IR, B	OR, B	IR, OR, B	IR, B
New bearing at the start (log)	yes	yes	yes	yes
Reviewer's verdict on the healthy labels	supported	supported	supported	**partly contradicted** (step at M0061)

Label provenance:

- The Run 3 label table is `run3/prep/label_table_run3.json`, SHA-256 `eb1e8e8e…`.
- Its B01–B04 part reproduces the Runs 1–2 table hash (`4eff40d8…`) exactly, which proves the labels of the earlier runs are unchanged.
- The manual two-person sign-off was not required (team decision). The labels remain **provisional** under assumptions A1–A4: end-stage proxy, defect caused the stop, not an onset time, healthy start.

6. Feature Extraction and Preprocessing

1. **Per measurement** (`paderborn.py`, unchanged): the rear accelerometer at 42.67 kHz is converted to g (10 g/V), and its RMS, peak, kurtosis and crest factor are computed; band energies 1–5 kHz and 5–20 kHz come from the full-rate spectrum.
2. **Per bearing** (`mtl_inputs.py`, unchanged):
   - Warm-up rows are removed.
   - On the first 100 operating measurements, each feature is fitted as f = b0 + b1 · ln rpm.
   - z = (f − fit) / σ.
   - Speed gives x7.
   - A first-order thermal model gives the temperature residual x8, which is 0 until the model is calibrated, and 0 throughout if the bearing fails the thermal validity rule (B08).
3. **Run 3's two input variants (the ARM computes them, so the hardware is unchanged):**
   - **clip16** (Runs 1–2): x = clip(z, −16, +16). INT8 step 16/127 = 0.126.
   - **slog** (candidates C and E): x = sign(z) · ln(1 + min(|z|, 10,000)). This keeps the order of values beyond 16 instead of flattening them, and is almost linear near 0. INT8 step ln(10,001)/127 = 0.0725.
4. **Temperature variants:** x8 as computed (A, B, C), or x8 = 0 on every row in training and in use (D, E).

No measurement of a validation or held-out bearing is used to fit anything shared across bearings. Each bearing is normalised on its own first 100 measurements, as a newly installed unit would commission itself. This has one consequence to keep in mind: the healthy labels are exactly that commissioning window, so their z-inputs are near 0 by construction.

7. Training Run 3 Configuration

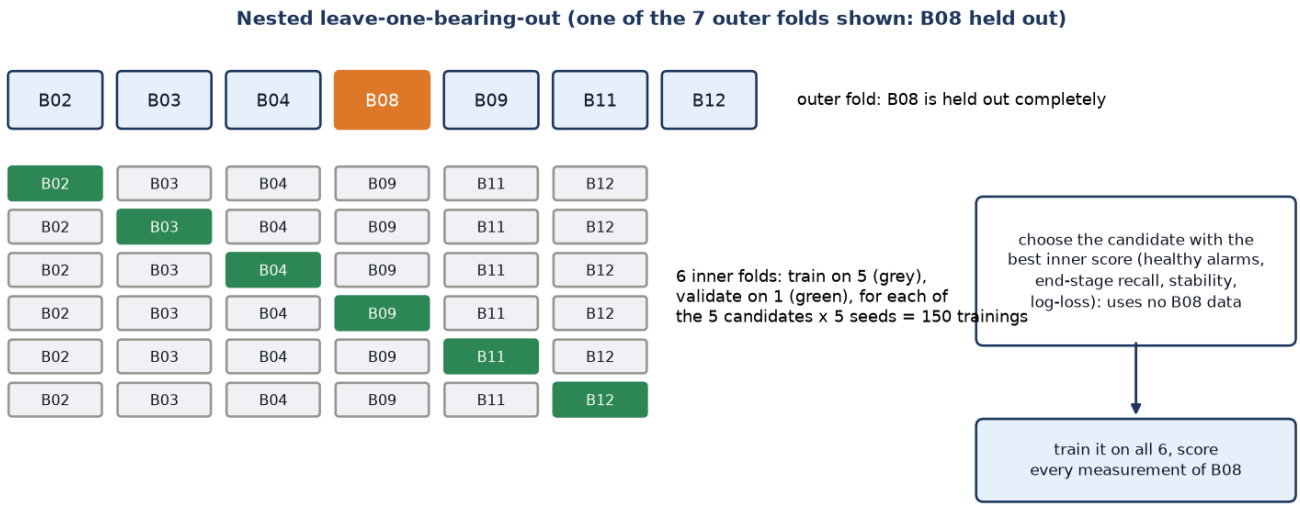
7.1 Candidates (declared in `train_deterioration_run3.py` and written to `train/DESIGN.json` before any training)

Candidate	Weight decay	Inputs	Temperature x8	Purpose
A: Run 1–2 settings	0	clip16	used	baseline; reproduces Run 1 bit for bit on Run 1's folds
B: weight decay 0.01	0.01	clip16	used	**pre-declared primary** (Run 2's suggestion; exploratory for B02–B04)
C: B + log inputs	0.01	slog	used	does the ±16 clip hide information?
D: B, no temperature	0.01	clip16	0	does the temperature shortcut matter?
E: C + D	0.01	slog	0	both

Everything else is Run 1–2's loop:

- Adam, learning rate 0.001, 2,000 steps;
- batches of 64 rows, half healthy and half end-stage (class balancing by sampling);
- temperature dropout 50 %;
- masked binary cross-entropy loss;
- the health head frozen;
- seeds 0–4 (Runs 1–2 used 0–2);
- threshold 0.5, fixed and never tuned;
- the alarm latched after 3 consecutive scores above the threshold.

7.2 The validation protocol: nested leave-one-bearing-out



- **Outer loop.** Each of the 7 bearings is held out in turn. The network is trained on the other 6 and scored on every measurement of the held-out bearing.
- **Inner loop (model selection).** For each outer bearing, every candidate is trained on 5 of the remaining 6 and validated on the 6th, six times over. The candidate with the best inner score is chosen, then judged on the outer bearing. 7 × 6 × 5 × 5 = 1,050 inner trainings.
- **Selection rule (fixed in advance).** A lexicographic order on the means:
  1. fewest healthy-reference rows alarmed;

2. highest end-stage recall;
3. fewest alarm-state changes of the **unlatched** alarm (stability, which needs no labels);
4. lowest log-loss on the labelled rows.

Alarm timing is never used, because the onset of damage is unknown.

- **Final model.** The same rule applied to the 7-fold outer results picks the deployment candidate, which is then trained on all 7 bearings (5 seeds). No held-out data is left for it, so the nested results are its performance estimate.

Leakage controls (in the code):

- `splits.assert_development` runs on every training set.
- Inner folds exclude the outer bearing.
- Every outer model is checked to have been trained on exactly 6 bearings, none of them the held-out one.
- INT8 calibration uses the training bearings only.
- The locked test bearings B06, B07, B13 and B17 are never loaded.

Exploratory versus unbiased (corrected after the independent review).

The whole candidate set was shaped by Run 2's held-out results on B02–B04:

- weight decay 0.01 was picked there;
- the log inputs answer Run 2's finding that every end-stage row sits at the clip;
- the no-temperature option answers Run 2's x8 check.

The latch was adopted after Run 1's flicker on B02–B04 (introduced in Run 2); the 0.5 threshold and the 50 % temperature dropout were fixed in Run 1's design. The thermal-model thresholds were drawn from B01 and B03's temperature curves.

So B02–B04 are exploratory folds for every candidate, A included (A is scored with the latch adopted after Run 1, and its temperature input uses thresholds drawn from B03), and only **B08, B09, B11 and B12 give unbiased estimates.** The results are reported for both groups (section 11).

The Run 3 script and `DESIGN.json` were written after the new bearings' preprocessing summaries had been printed (clipping counts, temperature masking, detector alarm ids), but before any network was trained on them and before any labelled-row outcome was inspected. The candidates themselves come from Run 2 and from the requested study list: weight decay, clipping, temperature.

7.3 Independent review of the protocol

Before the results were trusted, three independent reviewers (separate agents) read the code and the saved outputs, each with a different focus: leakage, correctness and evaluation design. Every serious finding was then checked again by a separate agent asked to refute it.

Confirmed clean:

- inner folds exclude the outer bearing;
- the selection for an outer bearing uses only its inner results (recomputed on one case);
- the validation-loss curve does not change the weights (bit-identical with and without it);
- INT8 hidden scales use training bearings only;
- no locked test bearing was loaded or even processed.

Confirmed problems, all handled in this report:

- the "unbiased" label must be restricted to the four new bearings;
- the labelled-row metrics are met by construction;
- the selection tie;
- B12's commissioning step;
- the temperature confound;
- the unfair parts of the reference comparison;

- the missing constant-speed regime;
- the v1 RTL row sampling;
- the INT8 dead-neuron overflow.

Minor code items (an unenforced reproduction assert, a missing digest recomputation, no Z-cap define in the C header) were checked by hand. The reproduction and the label digest both hold, and the npz labels match the label table exactly.

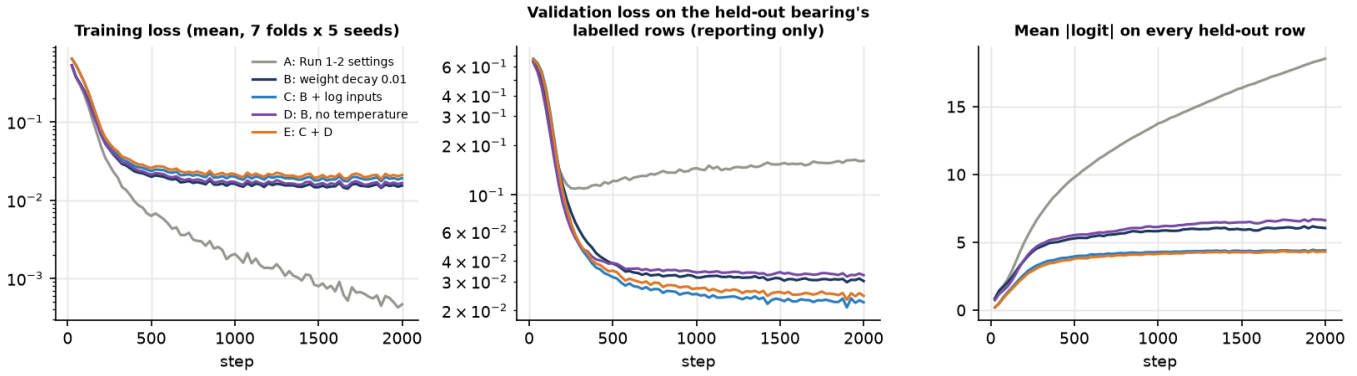
7.4 Compute

Training, evaluation and the first INT8/RTL pass ran in 119 s on 8 processes. Training is deterministic: as a check, the seed-0 network of each Run 1 fold was reproduced bit for bit (`run1_reproduced`: B02, B03, B04 all true).

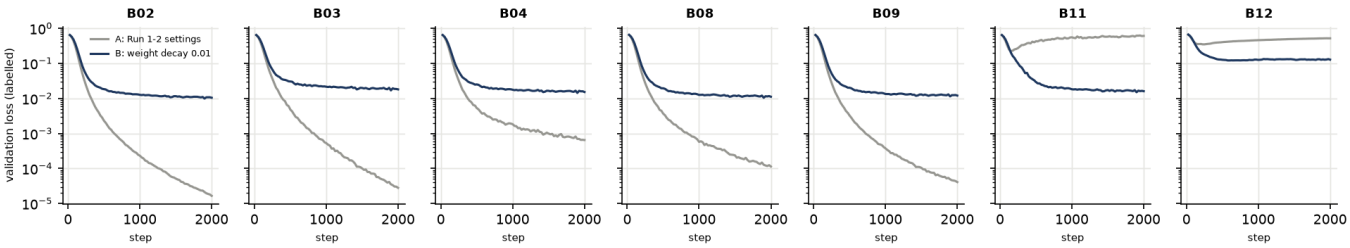
8. Improvements and Changes from Run 2

Change	Why	Effect (details in later sections)
4 more bearings (3 → 7), 2 new load and speed regimes, 64 kHz	Runs 1–2 saw only one regime	the first test of generalisation across regimes; B12 exposes a failure mode
Nested leave-one-bearing-out, 5 seeds	to choose a configuration without biasing its estimate	an estimate of the whole selection procedure, unbiased on the four new bearings (B02–B04 are exploratory, section 7.2)
Weight decay 0.01	Run 2's over-confidence diagnosis	logits about 3 times smaller; flicker reduced on B02, B03, B08 and B04 (65–75 → 7–35 changes, reduced but not removed), but **not on B11** (12–108 → 69–103)
Log-compressed inputs	the ±16 clip flattens gradual deterioration	**the largest relative stability gain** (mean unlatched changes 15.5 → 2.9, –81 %, against –49 % for weight decay: 30.3 → 15.5)
Temperature input off (option)	the x8 = 0 shortcut, B11's T1 sensor fault, B08's masked x8	no loss in healthy alarms (0) or bearings detected (6 of 7); without log inputs flicker rises (D against B: 15.5 → 20.7), and first alarms come later on B09 and B11; removes the shortcut and the sensor-fault dependence
INT8 export: dead-neuron fix	weight decay drove some weights to ~1e-35, overflowing the INT32 bias	all biases now fit 32 bits; FP32 output unchanged (difference 0)
Prototype detector replaced as the reference	it cannot run at 64 kHz	(A, F42, speed) on all 7 bearings

9. Training Results and Learning Curves



- **Training loss.** Without weight decay (A), the training loss falls to 0.0005. (Run 3's "final training loss" is the mean of the last 25 steps; Runs 1–2 stored the last step's loss.) With weight decay (B–E) it levels off at 0.015–0.021, because the penalty stops the weights from growing without bound.
- **Validation loss** on the held-out bearing's labelled rows (recorded every 25 steps, **never used to stop training or to choose anything**):
  - A reaches its minimum of about 0.11 near step 300, then **rises** to about 0.16 while its training loss keeps falling. The rise comes from B11 and B12, where A becomes confidently wrong: its end-stage scores fall towards 0. On the other five bearings A's validation loss keeps falling. That is over-fitting into over-confidence.
  - B–E settle at 0.022–0.033.
- **Mean |logit| on the held-out rows** grows to about 19 for A but stays at 4–7 for B–E. The scores of the regularised models are much less extreme.



Candidate	Healthy rows alarmed (35 runs)	End-stage recall	Unlatched changes (mean)	Labelled log-loss	Highest score on a healthy row	Scores < 0.001 or > 0.999	Unlabelled scores in 0.05–0.95	Final training loss	Chosen by nested selection (of 7 folds)
A: Run 1-2 settings	0	85.7 %	30.3	0.1610	0.225	88.9 %	3.6 %	0.0005	0
B: weight decay 0.01	0	85.7 %	15.5	0.0302	0.358	37.5 %	18.7 %	0.0154	0
C: B + log inputs	0	85.7 %	2.9	0.0224	0.476	3.4 %	17.7 %	0.0191	4
D: B, no temperature	0	85.7 %	20.7	0.0327	0.408	37.9 %	9.0 %	0.0166	0
E: C + D	0	85.7 %	2.8	0.0245	0.542	0.8 %	19.8 %	0.0210	3

10. Per-Bearing Validation Results

Each table gives, for one held-out bearing, the selected configuration E, the nested choice for that fold, the primary B and the Run 1–2 settings A (ranges over 5 seeds), and the references.

- **Healthy rows above / alarmed:** for the MLP rows, the count over 5 seeds (500 rows) of raw scores above 0.5, and of rows with the latched alarm on. The anomaly detector and B02's 6.5 g limit have one run (100 rows). The detector's threshold is the 99th percentile of these same 100 commissioning rows, so its "1 above" holds by construction.
- **Changes (unlatched):** the alarm-state changes of the unlatched alarm (on while the last 3 scores are above 0.5, Run 1's rule), a measure of stability; the deployed latched alarm changes at most once.

10.1 B02 (held out; exploratory fold)

Detector	First alarm (id)	Clock time	Min. before last record	Changes (latched)	Changes (unlatched)	Healthy rows above / alarmed	End-stage recall	Unlabelled rows alarmed
MLP E (selected), latched	867–868	13:16:16–13:16:33	80.0–80.3	1	1	0 / 0	100 %	25.5 %
MLP nested choice (E), latched	867–868	13:16:16–13:16:33	80.0–80.3	1	1	0 / 0	100 %	25.5 %
MLP B (primary, weight decay), latched	865–867	13:15:35–13:16:16	80.3–81.0	1	1	0 / 0	100 %	25.6 %
MLP A (Run 1–2 settings), latched	611–906	11:58:28–13:28:01	68.6–158.1	1	19–21	0 / 0	100 %	29.1 %
Anomaly detector (A, F42, speed)	865	13:15:35	81.0	1	1	1 / 0	100 %	25.7 %
Anomaly detector (A, F128, speed)	864	13:15:16	81.3	1	1	1 / 0	100 %	25.8 %
B02's 6.5 g limit on the rear sensor, latched	1109	14:34:28	2.1	1	3	0 / 0	100 %	0.3 %

10.2 B03 (held out; exploratory fold; temperature-stopped)

Detector	First alarm (id)	Clock time	Min. before last record	Changes (latched)	Changes (unlatched)	Healthy rows above / alarmed	End-stage recall	Unlabelled rows alarmed
MLP E (selected), latched	561	18:07:08	16.5	1	1	0 / 0	100 %	9.6 %
MLP nested choice (C), latched	561	18:07:08	16.5	1	1	0 / 0	100 %	9.6 %
MLP B (primary, weight decay), latched	558–561	18:06:11–18:07:08	16.5–17.5	1	1	0 / 0	100 %	10.0 %
MLP A (Run 1–2 settings), latched	580–590	18:13:09–18:16:15	7.4–10.5	1	3–5	0 / 0	100 %	4.3 %
Anomaly detector (A, F42, speed)	549	18:03:16	20.4	1	1	1 / 0	100 %	12.0 %
Anomaly detector (A, F128, speed)	549	18:03:16	20.4	1	1	1 / 0	100 %	12.0 %
B02's 6.5 g limit on the rear sensor, latched	never	–	–	0	0	0 / 0	0 %	0.0 %

10.3 B04 (held out; exploratory fold)

Detector	First alarm (id)	Clock time	Min. before last record	Changes (latched)	Changes (unlatched)	Healthy rows above / alarmed	End-stage recall	Unlabelled rows alarmed
MLP E (selected), latched	445	14:58:24	208.6	1	1	0 / 0	100 %	69.3 %
MLP nested choice (C), latched	444–445	14:58:06–14:58:24	208.6–208.9	1	1	0 / 0	100 %	69.4 %
MLP B (primary, weight decay), latched	444	14:58:06	208.9	1	7–35	0 / 0	100 %	69.4 %
MLP A (Run 1–2 settings), latched	489	15:12:25	194.6	1	65–75	0 / 0	100 %	64.8 %
Anomaly detector (A, F42, speed)	442	14:57:25	209.6	1	1	1 / 0	100 %	69.7 %
Anomaly detector (A, F128, speed)	442	14:57:25	209.6	1	1	1 / 0	100 %	69.7 %
B02's 6.5 g limit on the rear sensor, latched	1113	18:26:40	0.3	1	1	0 / 0	40 %	0.0 %

10.4 B08 (new; unbiased fold; temperature-stopped)

Detector	First alarm (id)	Clock time	Min. before last record	Changes (latched)	Changes (unlatched)	Healthy rows above / alarmed	End-stage recall	Unlabelled rows alarmed
MLP E (selected), latched	1216	21:46:50	214.1	1	1	0 / 0	100 %	36.3 %
MLP nested choice (E), latched	1216	21:46:50	214.1	1	1	0 / 0	100 %	36.3 %
MLP B (primary, weight decay), latched	1215–1216	21:46:16–21:46:50	214.1–214.7	1	1–3	0 / 0	100 %	36.3 %
MLP A (Run 1–2 settings), latched	1215–1219	21:46:16–21:47:36	213.4–214.7	1	27–47	0 / 0	100 %	36.2 %
Anomaly detector (A, F42, speed)	614	18:16:28	424.5	1	7	1 / 0	100 %	72.3 %
Anomaly detector (A, F128, speed)	614	18:16:28	424.5	1	7	1 / 0	100 %	72.3 %
B02's 6.5 g limit on the rear sensor, latched	1674	00:26:12	54.8	1	2	0 / 0	100 %	8.9 %

10.5 B09 (new; unbiased fold)

Detector	First alarm (id)	Clock time	Min. before last record	Changes (latched)	Changes (unlatched)	Healthy rows above / alarmed	End-stage recall	Unlabelled rows alarmed
MLP E (selected), latched	226–230	17:14:14–17:15:46	644.0–645.5	1	5–7	0 / 0	100 %	95.4 %
MLP nested choice (E), latched	226–230	17:14:14–17:15:46	644.0–645.5	1	5–7	0 / 0	100 %	95.4 %
MLP B (primary, weight decay), latched	202	17:04:43	655.0	1	5–11	0 / 0	100 %	96.9 %
MLP A (Run 1–2 settings), latched	207–212	17:06:51–17:08:28	651.3–652.9	1	7–13	0 / 0	100 %	96.6 %
Anomaly detector (A, F42, speed)	211	17:08:04	651.7	1	9	1 / 0	100 %	96.4 %
Anomaly detector (A, F128, speed)	207	17:06:51	652.9	1	3	1 / 0	100 %	96.6 %
B02's 6.5 g limit on the rear sensor, latched	never	–	–	0	0	0 / 0	0 %	0.0 %

10.6 B11 (new; unbiased fold; 64 kHz)

Detector	First alarm (id)	Clock time	Min. before last record	Changes (latched)	Changes (unlatched)	Healthy rows above / alarmed	End-stage recall	Unlabelled rows alarmed
MLP E (selected), latched	712–713	12:38:54–12:39:08	337.9–338.1	1	1–21	3 / 0	100 %	66.9 %
MLP nested choice (C), latched	707–711	12:37:35–12:38:39	338.4–339.4	1	3–11	0 / 0	100 %	67.1 %
MLP B (primary, weight decay), latched	590–699	12:06:36–12:35:24	341.6–370.4	1	69–103	0 / 0	100 %	72.4 %
MLP A (Run 1–2 settings), latched	540–1877	11:53:12–17:56:17	20.7–383.8	1	12–108	0 / 0	100 %	58.0 %
Anomaly detector (A, F42, speed)	706	12:37:18	339.7	1	1	1 / 0	100 %	67.3 %
Anomaly detector (A, F128, speed)	not available (no 20–64 kHz band at 64 kHz)	–	–	–	–	–	–	–
B02's 6.5 g limit on the rear sensor, latched	never	–	–	0	0	0 / 0	0 %	0.0 %

10.7 B12 (new; unbiased fold; 64 kHz)

Detector	First alarm (id)	Clock time	Min. before last record	Changes (latched)	Changes (unlatched)	Healthy rows above / alarmed	End-stage recall	Unlabelled rows alarmed
MLP E (selected), latched	never	–	never	0	0	0 / 0	0 %	0.0 %
MLP nested choice (C), latched	never	–	never	0	0	0 / 0	0 %	0.0 %
MLP B (primary, weight decay), latched	never	–	never	0	0	0 / 0	0 %	0.0 %
MLP A (Run 1–2 settings), latched	never	–	never	0	0	0 / 0	0 %	0.0 %
Anomaly detector (A, F42, speed)	563	10:41:34	40.8	1	14	1 / 0	100 %	30.2 %
Anomaly detector (A, F128, speed)	not available (no 20–64 kHz band at 64 kHz)	–	–	–	–	–	–	–
B02's 6.5 g limit on the rear sensor, latched	never	–	–	0	0	0 / 0	0 %	0.0 %

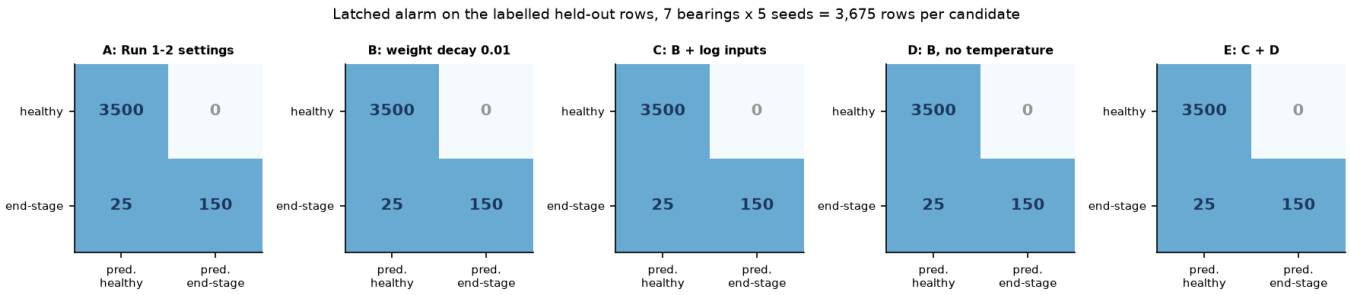
Variation across seeds and bearings.

- With E, the first alarm differs by at most 4 measurements between the 5 seeds on every bearing.
- With A, it differs by up to 295 measurements on B02 and 1,337 on B11 (but by 0–10 on B03, B04, B08 and B09). The unregularised network is not reliably reproducible across seeds.

11. Accuracy, Precision, Recall, F1 and Confusion Matrices

These scores are computed on the labelled held-out rows only: 100 healthy and 5 end-stage rows per bearing, which is 3,675 rows per candidate over 7 bearings × 5 seeds. The anomaly detector and B02's 6.5 g limit have one run per bearing (735 rows).

Model	Decision	Accuracy	Precision	Recall	F1	TN / FP / FN / TP
A: Run 1-2 settings	latched alarm	0.9932	1.0000	0.8571	0.9231	3500 / 0 / 25 / 150
A: Run 1-2 settings	raw score > 0.5	0.9875	1.0000	0.7371	0.8487	3500 / 0 / 46 / 129
B: weight decay 0.01	latched alarm	0.9932	1.0000	0.8571	0.9231	3500 / 0 / 25 / 150
B: weight decay 0.01	raw score > 0.5	0.9932	1.0000	0.8571	0.9231	3500 / 0 / 25 / 150
C: B + log inputs	latched alarm	0.9932	1.0000	0.8571	0.9231	3500 / 0 / 25 / 150
C: B + log inputs	raw score > 0.5	0.9943	1.0000	0.8800	0.9362	3500 / 0 / 21 / 154
D: B, no temperature	latched alarm	0.9932	1.0000	0.8571	0.9231	3500 / 0 / 25 / 150
D: B, no temperature	raw score > 0.5	0.9932	1.0000	0.8571	0.9231	3500 / 0 / 25 / 150
E: C + D	latched alarm	0.9932	1.0000	0.8571	0.9231	3500 / 0 / 25 / 150
E: C + D	raw score > 0.5	0.9924	0.9804	0.8571	0.9146	3497 / 3 / 25 / 150
Nested selection (unbiased)	latched alarm	0.9932	1.0000	0.8571	0.9231	3500 / 0 / 25 / 150
Anomaly detector (A, F42, speed) (7 bearings, 1 run)	latched alarm	1.0000	1.0000	1.0000	1.0000	700 / 0 / 0 / 35
B02's 6.5 g limit on the rear sensor	latched alarm	0.9687	1.0000	0.3429	0.5106	700 / 0 / 23 / 12



These row-level numbers carry little information. Two trivial rules, which use no learning and no data, show why:

Trivial rule (no learning, no data)	End stage latched before the stop	Healthy rows alarmed	Accuracy on labelled rows	First alarm, min before last record
Never alarm	0 of 7	0	0.9524	–
Latch an alarm at the first measurement after commissioning (row 101)	7 of 7	0	1.0000	133–677

- Healthy rows can only be falsely alarmed inside the commissioning window, where the inputs are near 0 by construction.
- For the latched MLP alarms and the anomaly detector, all end-stage rows of a bearing are caught or missed together, by one latch. (B02's 6.5 g limit latches inside B04's end stage and catches 2 of its 5 rows.)

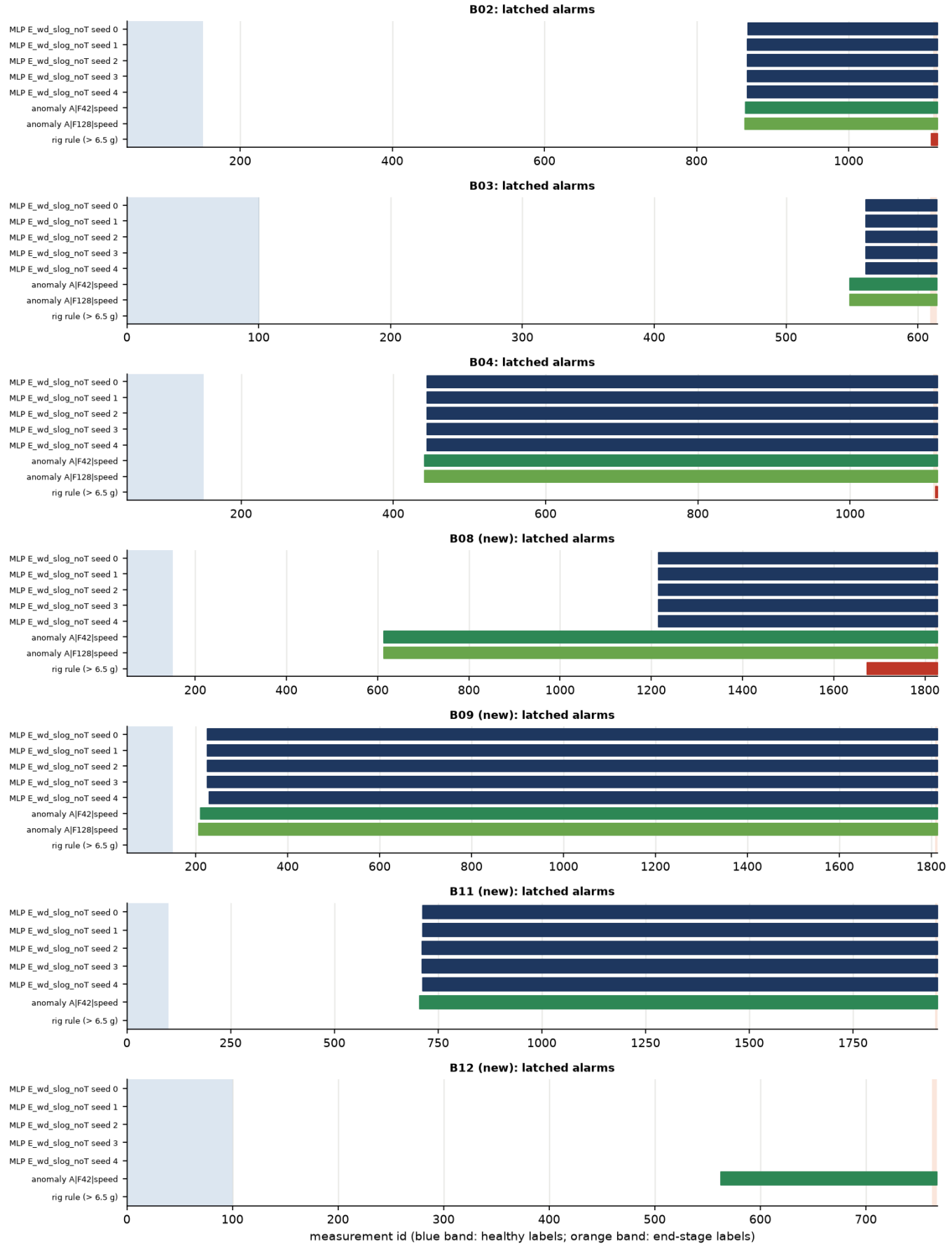
The evidence that matters is per bearing. The table below gives exact (Clopper–Pearson) 95 % intervals; seeds do not add independent evidence, because every candidate gives the same outcome on a bearing for all 5 seeds.

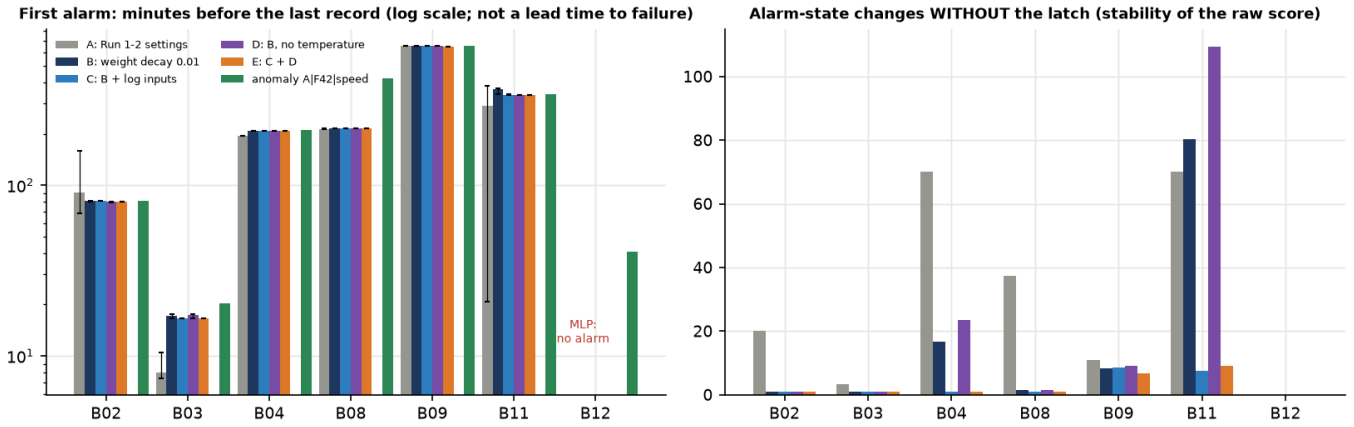
Bearings	E (selected)	C	B (primary)	A (Run 1–2 settings)	Nested selection	Anomaly detector	Trivial rule: latch right after commissioning
all 7	6/7 (0.42–1.00)	6/7 (0.42–1.00)	6/7 (0.42–1.00)	6/7 (0.42–1.00)	6/7 (0.42–1.00)	7/7 (0.59–1.00)	7/7 (0.59–1.00)
exploratory (B02–B04)	3/3 (0.29–1.00)	3/3 (0.29–1.00)	3/3 (0.29–1.00)	3/3 (0.29–1.00)	3/3 (0.29–1.00)	3/3 (0.29–1.00)	3/3 (0.29–1.00)
unbiased new (B08, B09, B11, B12)	3/4 (0.19–0.99)	3/4 (0.19–0.99)	3/4 (0.19–0.99)	3/4 (0.19–0.99)	3/4 (0.19–0.99)	4/4 (0.40–1.00)	4/4 (0.40–1.00)
all except B12 (descriptive)	6/6 (0.54–1.00)	6/6 (0.54–1.00)	6/6 (0.54–1.00)	6/6 (0.54–1.00)	6/6 (0.54–1.00)	6/6 (0.54–1.00)	6/6 (0.54–1.00)

How to read the row-level numbers above:

- **With the latched alarm, all 25 false negatives for every MLP candidate are B12's end-stage rows** (5 rows × 5 seeds), and every other bearing's end stage is caught. With the raw score > 0.5, C misses 21 (one B12 end-stage row scores above 0.5 in 4 of 5 seeds) and A misses 46 (B12's 25, plus 21 on B11 where its score falls back below 0.5).
- **The latched alarm has no false positives.** With the raw score > 0.5, E has 3: the same B11 healthy row (M0073) in seeds 1–3, a single score of 0.51–0.54 that is never 3 in a row.
- **The scores are dominated by the easiest rows** (brand-new against worn-out), so they are not an estimate of accuracy in service.
- **The anomaly detector scores higher** (recall 1.0) because it also catches B12.

12. Early Warning and Alarm-Timing Comparison





Bearing	Selected E: first alarm (min before last record)	Anomaly detector (A, F42, speed)	Earlier detector, by	Unbiased fold?
B02	80.0–80.3	81.0	anomaly detector, 0.7–1.0 min	no
B03	16.5	20.4	anomaly detector, 3.9 min	no
B04	208.6	209.6	anomaly detector, 1.0 min	no
B08	214.1	424.5	anomaly detector, 210 min	yes
B09	644.0–645.5	651.7	anomaly detector, 6–8 min	yes
B11	337.9–338.1	339.7	anomaly detector, 1.6–1.8 min	yes
B12	never	40.8	anomaly detector (MLP misses it)	yes

The same comparison in measurements (one measurement is about 12–23 s):

Bearing	Anomaly detector first alarm (id)	Detector, min before last record	E: lag (measurements; + = later)	C: lag	B: lag
B02	865	81.0	+2, +3	+1	+0, +2
B03	549	20.4	+12	+12	+9, +12
B04	442	209.6	+3	+2, +3	+2
B08	614	424.5	+602	+602	+601, +602
B09	211	651.7	+15, +19	-4	-9
B11	706	339.7	+6, +7	+1, +3, +5	-116, -7
B12	563	40.8	never	never	never

On the new bearings specifically, the selected network never warns earlier than the anomaly detector (nor on B02–B04). On B02, B03, B04 and B11 its first alarm comes 2–12 measurements after the detector's, essentially the same event.

All four non-selected candidates raise an earlier alarm on a new bearing for at least one seed, but none is a usable early warning:

- **B (weight decay only)** alarms at M0202 on B09, 3.3 min before the detector, and 2–31 min earlier on B11. On B11 its unlatched alarm flickers 69–103 times, and with x8 = 0 those B11 alarms move 15–124 measurements later: they are driven by the temperature input, not by vibration (x8 is +12.7 to +13.8 σ there, before the T1 fault).
- **C (log inputs with temperature)** alarms at M0207 on B09: 1.2 min before the (A, F42, speed) detector and at the same measurement as the original F128 prototype. With x8 = 0 it moves 23 measurements later, so this small lead also

comes from temperature.

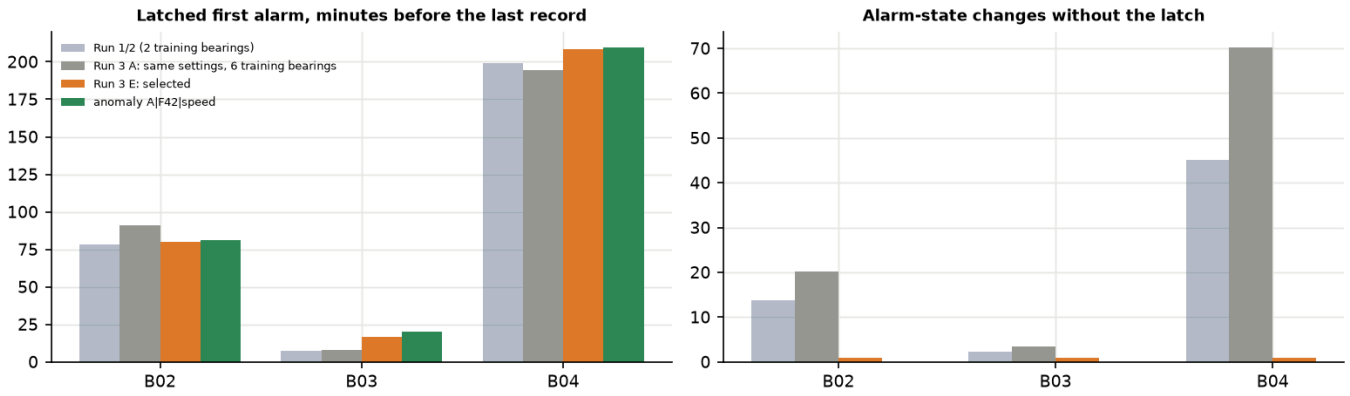
- **D (B, no temperature)** alarms at M0207 on B09 in 2 of 5 seeds (the others at M0211, the detector's measurement).
- **A** raises some very early alarms (158 min before the end on B02, 384 min on B11) and alarms at M0207 on B09 in 4 of 5 seeds, but its first alarms differ by hundreds of measurements between seeds.

On B08 the anomaly detector alarms 7 hours before the end, at M0614, on a brief excursion of its score (2.03, 2.45 and 2.13 at M0612–M0614 against a threshold of 1.87) that is over by M0615, long before the large step that the MLP waits for (M1216). Neither time can be verified as an onset.

13. Run 1 vs. Run 2 vs. Run 3 Comparison

Runs 1 and 2 evaluated only B02–B04 and trained each fold on 2 bearings; Run 3 trains each fold on 6. Each cell gives the first alarm in minutes before the last record, then the alarm-state changes of the same unlatched rule (on while the last 3 scores are above 0.5), so the counts are like-for-like; the deployed latched alarm changes once.

Bearing	Run 1 (unlatched rule; 2 training bearings, 3 seeds)	Run 2 (latched; same networks)	Run 3 A (Run 1–2 settings, 6 training bearings, 5 seeds; unlatched changes)	Run 3 E (selected; unlatched changes)	Anomaly detector (A, F42, speed)
B02	76.6–79.5 / 9–17	76.6–79.5 / 1	68.6–158.1 / 19–21	80.0–80.3 / 1	81.0 / 1
B03	7.4 / 1–3	7.4 / 1	7.4–10.5 / 3–5	16.5 / 1	20.4 / 1
B04	194.6–204.4 / 43–47	194.6–204.4 / 1	194.6 / 65–75	208.6 / 1	209.6 / 1



What improved:

- **Stability:** 1 change on B02–B04 for E, against 9–17 (B02), 1–3 (B03) and 43–47 (B04) for Run 1.
- **Earlier first alarms:** on all three bearings, by 0.5–3.7 min on B02, 9.1 min on B03 and 4.2–14.0 min on B04.
- **Less over-confidence:** scores below 0.001 or above 0.999 fall from 87.1 % (Runs 1–2, B02–B04) to 1.1 % for E on the same bearings (0.8 % over all 7).
- **Reproducibility across seeds.**

What did not improve:

- **The MLP is still later than the anomaly detector.**
- **More data alone does not help.** The Run 1–2 settings trained on 6 bearings (Run 3 A) flicker *more* (19–21, 3–5 and 65–75 changes), and their first alarm varies more between seeds on B02 and B03.
- **A new failure appears:** B12.

Comparability. Runs 1–2 never saw B08–B12, so on those bearings Run 3 has no earlier run to compare with. Run 3's B02–B04 folds were trained on 6 bearings, so the comparison mixes the effect of more data with the effect of the new settings. Run 3 A separates the two.

14. Comparison Against the Original Anomaly Detector

Criterion	MLP (selected E)	Anomaly detector (A, F42, speed)
Needs labels	yes (healthy and end-stage)	no (fits each bearing's own first 100 measurements)
Healthy-reference false alarms	0	0 (1 row above threshold per bearing, in-sample)
End-stage recall	6 of 7 bearings (misses B12)	**7 of 7**
First alarm	later or equal on every bearing	**earlier or equal on every bearing**
Stability (alarm-state changes, unlatched)	mean 0–9 per bearing (0 on B12, which never alarms; up to 21 on one B11 seed)	1–14 per bearing (14 on B12, 9 on B09, 7 on B08)
Latched alarm changes	1 (none on B12)	1 on every bearing
Runs on 64 kHz data	yes	the F42 version yes; the original F128 prototype no
Hardware	8 × 8 array, 1.78 µs	ARM only (no network)

Verdict: the trained MLP does not outperform the anomaly detector. Under Run 3's own selection rule the detector would rank first: 0 healthy rows alarmed and 7 of 7 end stages latched, against 6 of 7. The MLP's first alarms coincide with the detector's on five bearings (2–19 measurements later), come about 3.5 hours later on B08 and never come on B12. Its one advantage is a steadier raw score on B08 and B09.

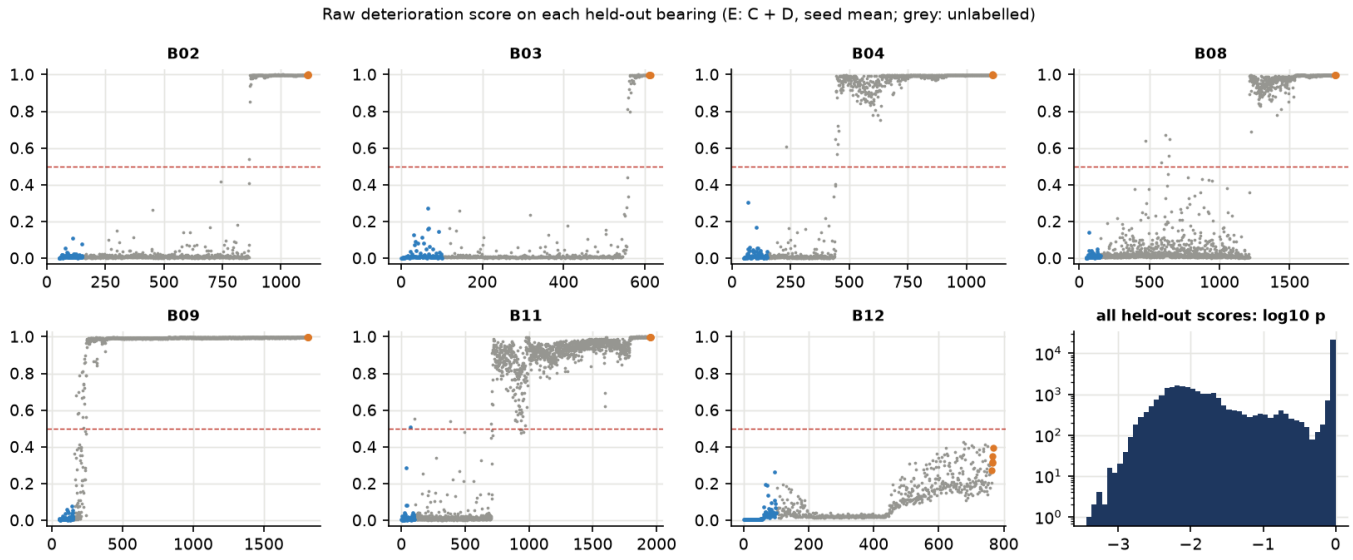
The reason is structural:

- The anomaly detector's threshold is the 99th percentile of the bearing's own healthy scores, so it flags any departure from normal.
- The network's decision boundary is learned from labels that are all extreme, so it waits for a large departure.
- Until labelled intermediate stages exist, a supervised network trained this way cannot be expected to warn earlier.

15. Prediction Confidence, Calibration and Input-Clipping Analysis

15.1 Over-confidence

- **Without weight decay** (A), 88.9 % of all held-out scores are below 0.001 or above 0.999, with a mean |logit| of 19.9.
- **With weight decay, log inputs and no temperature input** (E), that share is 0.8 % and the mean |logit| is 4.3 (C: 3.4 % and 4.4). Scores in the unlabelled middle spread out: 19.8 % fall between 0.05 and 0.95.
- **The price is a smaller safety margin on healthy rows.** E's highest healthy score is 0.54 (B11, against 0.23 for A). One B11 healthy row, M0073, went above 0.5 in three of the five seeds (0.51–0.54); it was a single spike, never three in a row, so no alarm was raised.



15.2 Calibration

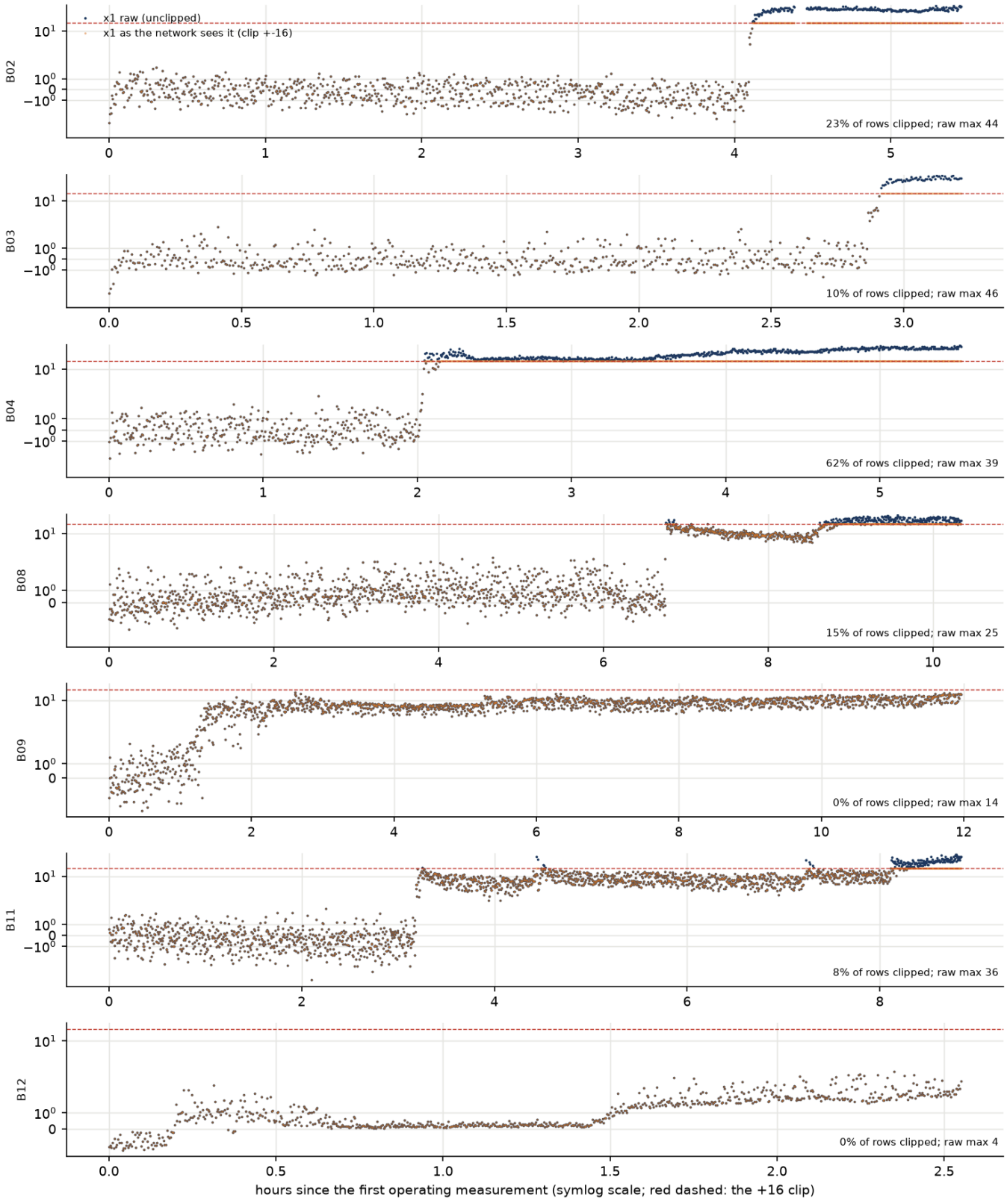
Calibration still cannot be verified.

- On the labelled rows the log-loss is low (0.025 for E).
- But those rows are only the two extremes, and the unlabelled middle has no ground truth.
- Class balancing (50 % end-stage in every batch, against 4.8 % in the labelled rows) also shifts the scores upward by design.

The output must be read as a **deterioration score with a threshold**, not as a probability of failure.

15.3 Does the ±16 clip hide information?

Yes. On B04 the ln-RMS input keeps rising from 16 to about 39 over the last 3 hours while the clipped network input stays at 16. Among the clipped rows, the Spearman rank correlation between the raw value and time is 0.83 for ln RMS and 0.86 for both band energies. B03 (0.67–0.73 for ln RMS and the 1–5 kHz band) and B11 (0.61–0.63) show the same. The share of rows with at least one input clipped is 10–64 % per bearing (table in section 5).

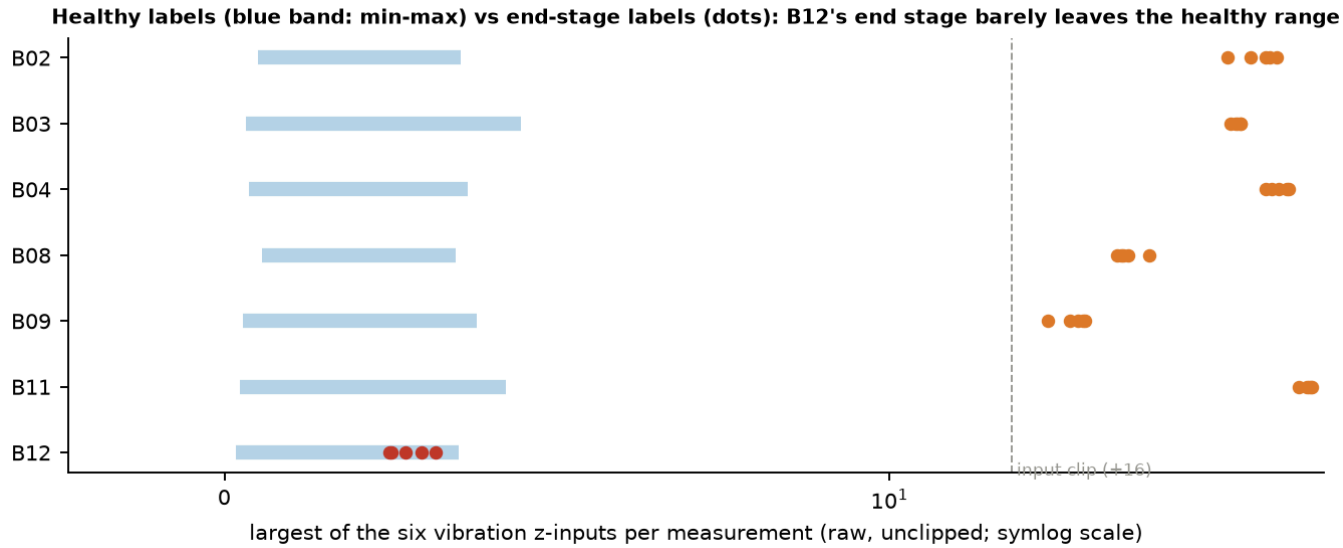


Effect of removing the clip (log-compressed inputs, C against B):

- the mean unlatched alarm-state changes fall from 15.5 to 2.9;
- B04's flicker goes from 7–35 changes to 1;
- the share of extreme scores falls from 37.5 % to 3.4 %.

In relative terms this is the largest single improvement in Run 3 (changes –81 %, extreme scores –91 %); in absolute terms adding weight decay (A → B) removed more (30.3 → 15.5 changes). The nested selection chose a log-input candidate on every one of the 7 outer folds.

15.4 Why B12 is missed



On every other bearing the end-stage rows reach a largest vibration z-input of 18–51. On B12 they reach only 2.5–3.2, inside B12's own healthy range (up to 3.5).

Two facts from the data review explain this:

1. B12's stop was most likely triggered by the **front** accelerometer (8.5 g against 4.9 g on the rear).
2. A **lasting step at M0061** inside B12's commissioning window inflates its reference spread.

A post-hoc check, explicitly not part of the evaluation (`run3_b12_diagnostic.py`): re-commissioning B12 on M0061–M0160 halves the spread (0.501 → 0.234). Even then the end-stage z reaches only 2.3–4.1, and the held-out E models alarm on just 2 of 5 seeds (18 min before the end). The anomaly detector alarms 40.8 min before the end either way.

15.5 The temperature shortcut

- **The shortcut.** x8 is exactly 0 on every healthy label (the thermal model is still calibrating) and non-zero at the end stage, so it separates the labels for a reason that has nothing to do with physics.
- **Data problems on top of it.** B11's T1 sensor fails later in the run, and B08's x8 is masked.
- **The evidence:**
  - Removing x8 (D against B, E against C) leaves recall and healthy false alarms unchanged and makes no consistent difference to stability (20.7 against 15.5 changes; 2.8 against 2.9).
  - Re-scoring the trained models with x8 = 0 shows where temperature does matter: B's early alarms on B11 move 15–124 measurements later, and C's alarm on B09 moves 23 measurements later. These earlier alarms are driven by the temperature residual, but not by a saturated input or the failed sensor: at B's B11 alarms x8 is +12.7 to +13.8 σ, below the clip, and T1 still agrees with T2 (x8 saturates only from M0724, the T1 fault starts about M0979).
- **The end-stage temperature values contradict each other across bearings,** so the temperature study is **inconclusive**:

Bearing	x8 on the 5 end-stage rows (raw)	x8 as the clip16 network sees it	x8 = 0 on all 100 healthy rows
B02	12.9, 12.8, 12.8, 12.9, 13.0	12.9, 12.8, 12.8, 12.9, 13.0	yes
B03	1.8, 1.9, 7.4, 10.5, 21.1	1.8, 1.9, 7.4, 10.5, 16.0	yes
B04	7.1, 7.4, 7.2, 7.2, 7.2	7.1, 7.4, 7.2, 7.2, 7.2	yes
B08	0.0, 0.0, 0.0, 0.0, 0.0	0.0, 0.0, 0.0, 0.0, 0.0	yes
B09	11.4, 11.1, 11.5, 11.0, 11.0	11.4, 11.1, 11.5, 11.0, 11.0	yes
B11	-59.5, -60.0, -59.5, -60.2, -60.6	-16.0, -16.0, -16.0, -16.0, -16.0	yes
B12	31.0, 31.2, 30.9, 30.8, 31.1	16.0, 16.0, 16.0, 16.0, 16.0	yes

B11's end-stage x8 is –60 σ, because its T1 sensor dropped out. B08's is 0 (masked). The others are +1.8 to +31 σ. A network cannot learn a consistent temperature signal from these 7 bearings.

- **Conclusion:** the temperature input currently adds no measurable value and carries two risks, so E's choice to ignore it is defensible, but the data cannot show whether temperature would help with a working sensor and a validity flag.
- **The team's call.** If the specification requires temperature, **C** is the equivalent alternative. It was trained and exported as well (section 16).

16. INT8 Quantization and RTL Verification

16.1 A bug found and fixed in the export path

- **The bug.** Weight decay drives the weights of some dead hidden neurons to about 1e-35. `export.build` then gives that output channel a scale of about 1e-37, so its bias quantizes to about 1e24, which overflows the INT32 bias and the cast.
- **The extent.** It affected 31 of 175 outer models and 2 of the 5 final E models (seeds 1 and 2). It did not change any INT8 decision in this run (the dead neurons output 0 either way): the first INT8 numbers in `train/results.json` are identical, row for row, to the fixed ones in `hardware/`, and none of the networks exported or RTL-simulated in v1 was affected. But the exported biases would not fit the hardware's 32-bit register, so the fix is required.
- **The fix** (`run3_hardware.py`, export path only; `export.py` and the trained weights are unchanged): weights with

w below 10^{-6} of their layer's largest are set to 0 before quantizing, and every hidden-layer bias is asserted to fit 32 bits (the output-layer bias, added by the ARM, is about

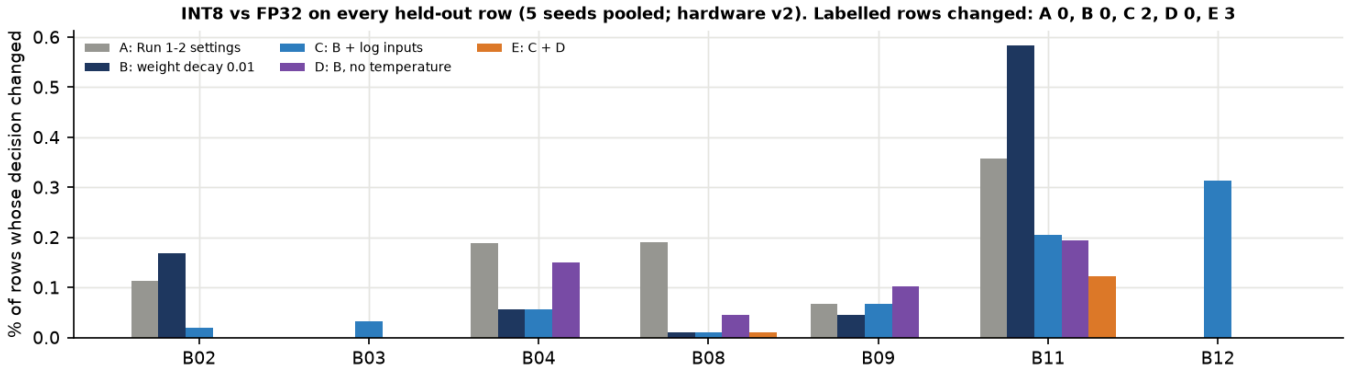
10^{3}).

- **Checked:** the FP32 output changes by **exactly 0** on every row. 136 of 175 outer models contained such dead weights.

16.2 FP32 against INT8 on every development bearing

Each model is checked on every row of its held-out bearing, through the bit-exact golden model (22-bit accumulator). Hidden scales come from the training bearings only.

Models	Networks	Rows	Decisions changed	Share	On labelled rows	Latched first alarm identical	Mean abs. logit error	Max abs. logit error	Networks with dead weights zeroed
A: Run 1-2 settings	35	45015	74	0.164 %	0	33	0.244	11.26	1
B: weight decay 0.01	35	45015	74	0.164 %	0	33	0.058	0.46	30
C: B + log inputs	35	45015	44	0.098 %	2	34	0.038	0.28	35
D: B, no temperature	35	45015	40	0.089 %	0	34	0.064	2.08	35
E: C + D	35	45015	13	0.029 %	3	35	0.039	0.35	35
Final E models (all 7 bearings, in-sample)	5	45015	39	0.087 %	4	34 of 35 bearing runs	0.041	0.29	5



- **E** changes 13 of 45,015 decisions (0.03 %), the fewest of all candidates, and its latched first alarm is identical in all 35 networks.
- **Labelled rows:** E changes 3 (one B11 healthy row in each of three seeds, each a score near 0.5).
- **C's final models** change 37 of 45,015 decisions (in-sample), none on a labelled row.

16.3 RTL simulation of the trained networks

`tb_mlp.sv` was run on the accelerator RTL, with `rtl_check.PARAMS` (P = 8, ACC = 22) and Icarus Verilog, against the golden model: every raw output sum, the last layer's buffer and every counter were compared.

Network	Batches of 64	Cycles per 64	M = 64	Cycles per 1	M = 1
Final E, seed 0 (all 7 bearings)	24	1642	PASS	382	PASS
Outer B_wd/B02 (seed 0)	9	1642	PASS	382	PASS
Outer B_wd/B03 (seed 0)	9	1642	PASS	382	PASS
Outer B_wd/B04 (seed 0)	9	1642	PASS	382	PASS
Outer B_wd/B08 (seed 0)	9	1642	PASS	382	PASS
Outer B_wd/B09 (seed 0)	9	1642	PASS	382	PASS
Outer B_wd/B11 (seed 0)	9	1642	PASS	382	PASS
Outer B_wd/B12 (seed 0)	9	1642	PASS	382	PASS
Outer C_wd_slog/B03 (seed 0)	9	1642	PASS	382	PASS
Outer C_wd_slog/B04 (seed 0)	9	1642	PASS	382	PASS
Outer C_wd_slog/B11 (seed 0)	9	1642	PASS	382	PASS
Outer C_wd_slog/B12 (seed 0)	9	1642	PASS	382	PASS
Outer E_wd_slog_noT/B02 (seed 0)	9	1642	PASS	382	PASS
Outer E_wd_slog_noT/B08 (seed 0)	9	1642	PASS	382	PASS
Outer E_wd_slog_noT/B09 (seed 0)	9	1642	PASS	382	PASS

The final C model (seed 0) also passed (24 batches of 64 and 10 single rows, 1,642 and 382 cycles). The cycle counts equal the hardware equation, `mlp_equation`, in every run.

The review noticed that the first RTL pass (v1) simulated only a prefix of each bearing's rows. The v2 runs above sample a spread of rows, and a supplementary run (`run3_supplement.py`) simulated **all 735 labelled rows, including all 35 end-stage rows**, for the seed-0 final E and C models (12 batches of 64, the last topped up): **PASS, bit-exact**.

Batch	Cycles	Latency at 214.286 MHz
1 measurement	382	**1.78 µs**
64 measurements	1,642	7.66 µs (0.12 µs per measurement)

The 8 × 8 array supports the network without any architecture change: 14 array passes, INT8 inputs, all accumulations fit 22 bits, both heads in one 16 × 2 layer. **These are simulation results; nothing in Run 3 ran on the ZedBoard.**

17. Current Limitations and Uncertainties

- **The bearing is the independent unit, and there are 7.** 6 of 7 latched (95 % interval 0.42–1.00); on the four unbiased bearings 3 of 4 (0.19–0.99). No claim of a detection rate is possible.
- **The labelled-row metrics are met by construction.** A trivial "latch after commissioning" rule matches or beats every candidate on them.
- **35 end-stage labels from 7 bearings.** Every one comes from the last 1–2 minutes of a test (48–114 s). The 30 from six bearings are at an extreme state (largest vibration z-input 18–51); B12's 5 reach only 2.5–3.2.
- **The labels are provisional and unreviewed.** B12's healthy window contains a vibration step, and for B11 and B12 the end labels rest on a stop probably triggered by the front sensor (the rear never reaches 6.5 g). On B12 the rear sensor barely registered it; on B11 it rose clearly (vibration z 48–51).
- **The onset of damage is unknown.** First-alarm times are not detection delays. "Earlier" is not verifiably "better".
- **The selected MLP does not beat the anomaly detector,** and it misses one of seven bearings.
- **In the final selection, the rule's first two criteria tied** for every candidate. The choice fell to stability, a label-free proxy, and E and C are within noise of each other (2.80 against 2.86 changes). In the nested selection, inner recall differed on B04, B11 and B12 and decided the pick (C) on B04 and B11.
- **The healthy-row margin is smaller** with weight decay (highest healthy score 0.54).
- **Scores are not calibrated probabilities.**
- **The 64 kHz bearings** are compared with the (A, F42, speed) detector, not the original F128 prototype.
- **The candidate set was designed from Run 2's held-out B02–B04 results** (and the thermal thresholds from B01/B03), so B02–B04's folds are exploratory for every candidate, A included (it is scored with the latch adopted after Run 1).
- **The constant-speed regime (B14–B17) is not covered** by any development bearing, but B17 is in the locked test set.
- **The 6.5 g reference is B02's logged limit applied to the rear sensor,** not the rig's actual rule on the new bearings, whose limit and channel are undocumented.
- **Only one accelerometer,** the rear one, while two stops were probably triggered by the front one.
- **Hardware results are simulation only.**
- **The Motor Health Index is untrained.**
- **The locked test bearings have not been evaluated.**

18. Recommended Improvements for Training Run 4

1. **Teach the network intermediate stages, or stop asking it to find them alone.** Options to evaluate under the same nested protocol:
   - Combine the two detectors: use the anomaly detector's per-bearing threshold as an input or a gate, and the MLP as a confirmer.
   - Extend the end-stage labels with the sensitivity windows already defined (END_N 10 or 20), declared in advance.
   - Express the MLP's score relative to its own commissioning distribution, as the anomaly detector does.
2. **Add a commissioning-validity check for vibration,** like the thermal V1–V3 rule. It would have flagged B12's step at M0061. Decide before Run 4 whether such bearings are re-commissioned or excluded.
3. **Decide the sensor question.** Two stops were probably triggered by the front accelerometer. Either confirm with the dataset authors which channel the limit used, or evaluate a front-sensor (or max-of-both) variant of the six vibration features, which keeps 8 inputs.
4. **Make the log-compressed inputs the default** (C or E), and move the dead-weight zeroing and the INT32 bias check into `export.py` itself.

5. **Temperature:** redesign x8 with an explicit validity flag and a T1/T2 consistency check (B11's T1 failure), or keep it disabled (E).
6. **More bearings:**
   - B10 (64 kHz, outer race), B14 and B15 (constant speed, the last regime not yet seen);
   - B05 and B16 only for their commissioning rows: both were found damaged at dismantling and lack end-stage labels only because B05 was interrupted and B16 has no stop criterion, so the rest of their recordings cannot be counted as healthy.
7. **Choose the threshold on the inner folds** rather than fixing it at 0.5. B12's end-stage scores of 0.25–0.41 show the fixed threshold matters.
8. **Freeze one model, then evaluate it once on B06, B07, B13 and B17,** and report the result whatever it is.

19. FPGA Deployment Readiness

Item	Status
Architecture fits the 8 × 8 array, INT8, 22-bit accumulator	**ready** (unchanged network; verified in RTL simulation)
Trained weights in the hardware format	**ready**: `export/MTL_run3_v2/seed0–4/` (E) and `export/MTL_run3_v2_C/seed0–4/` (C), each with W/Bias/Scl hex files, `mtl_run3.h` , `quant.json` and test vectors
Bit-exact RTL simulation of the trained network	**passed** (16 networks)
ARM-side code	to do: feature extraction, per-motor commissioning, the slog transform (one `log1p` per input), the latched alarm with a maintenance reset
Board test on the ZedBoard	**not done**: the trained network has never run on the board (the board passed 30/30 on-chip MLP tests with test models on 24 Oct, recorded on branch `Alex_edit`)
Model frozen and tested on the locked bearings	**not done**
Health-index output	**must stay disabled** (untrained)

20. Final Conclusion

What Run 3 accomplished:

- **Data:** four new run-to-failure bearings, from two new operating regimes, integrated with full integrity checking and a documented data review. That more than doubles the labelled development data.
- **Validation:** a leak-checked, nested leave-one-bearing-out evaluation of five configurations, reproducible bit for bit.
- **Stability:** log-compressed inputs with weight decay make the network much steadier and less over-confident, and bring its first alarm close to the anomaly detector's on five of seven bearings (about 3.5 h later on B08, never on B12).
- **Hardware:** the trained network passes INT8 and bit-exact RTL verification on the unchanged 8 × 8 accelerator, at 1.78 µs of accelerator compute per measurement (382 simulated clocks at 214.286 MHz; bus transfers and ARM work not included), after a real export bug was found and fixed.

What it did not achieve:

- **The network does not outperform the label-free anomaly detector.** It never warns earlier, and it misses B12.
- **The likely cause** lies in what the labels teach (only extreme end stages), in B12's data (a commissioning step, and a stop probably triggered by the other sensor) and in the fixed 0.5 threshold, not in the hardware: the FP32 models miss B12 too. The network size was never varied, so it is not ruled out.
- **Still open:** the Motor Health Index stays untrained, and every hardware result is from simulation (the detection results come from the recorded data, processed offline).

Next: combine the strengths of the two detectors, add a vibration commissioning check, settle the sensor question, and evaluate a frozen model once on the locked test bearings.

Files

What	Where
Data integration	`model/extract_paderborn_zips.py`, `model/newdata/<EXP>/` (not in git), `model/results/paderborn/B08`–`B12/` (processed CSV, `checks.json`, `extraction_manifest.json`, overview figure), `model/paderborn_experiments.json` (backup: `results/supervised/run3/provenance/`)
Preparation	`model/run3_prep.py` → `model/results/supervised/run3/prep/` (datasets, label table, folds, data quality)
Training	`model/train_deterioration_run3.py` → `model/results/supervised/run3/train/` (`DESIGN.json`, `results.json`, `curves.csv`, `inner_folds.csv`, `predictions/`, `models/` with 175 outer, 5 final E)
Hardware	`model/run3_hardware.py` → `results/supervised/run3/hardware/`, `export/MTL_run3_v2/`; `model/run3_final_c.py` → `train_alt_C/`, `hardware_alt_C/`, `export/MTL_run3_v2_C/`
Diagnostics	`model/run3_b12_diagnostic.py` → `results/supervised/run3/diagnostics/`
Figures and tables	`model/report/run3_figures.py`, `run3_tables.py`, `run3_diagrams.py` → `results/supervised/run3/analysis/`
Superseded (kept as a record)	`train/results.json` "int8" entries and `export/MTL_run3/` (before the dead-neuron fix)